Lab 5: Ingest Data using COPY INTO in Databricks

Lab Name

Ingest Data using COPY INTO in Databricks

Lab Summary

This lab demonstrates how to ingest data into a **Delta table using the COPY INTO command** in Databricks.

You will create a notebook, upload Parquet files into a volume, create a Delta table, and load the files using COPY INTO while supporting schema evolution.

Prerequisites

Before starting this lab, ensure:

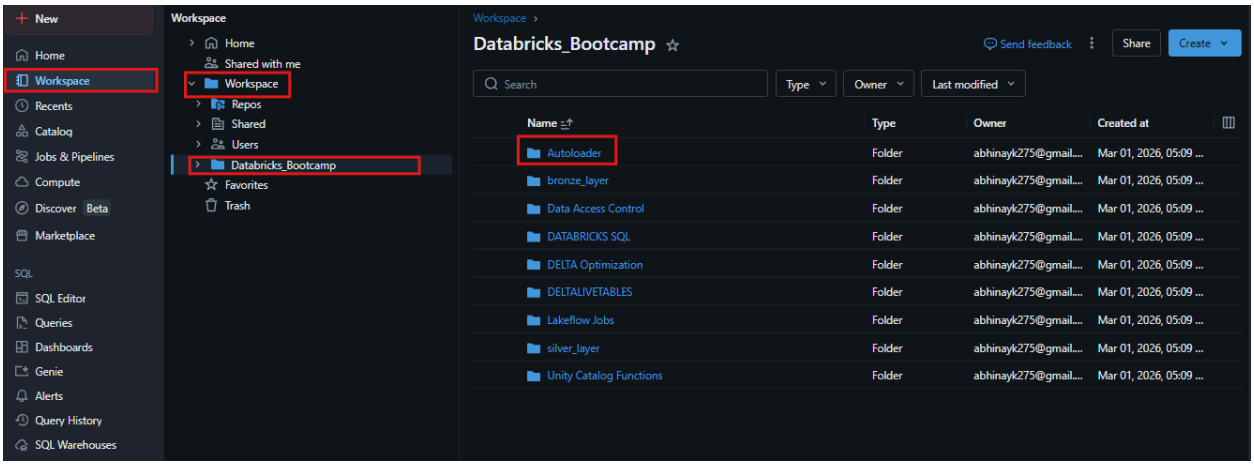
- Databricks workspace is available
- Serverless or cluster compute is running
- Access to **Databricks Bootcamp workspace**
- Parquet dataset files available
- Basic knowledge of:
  - Databricks workspace
  - Delta tables
  - SQL

Steps to Complete the Lab

Task 1: Create Notebook

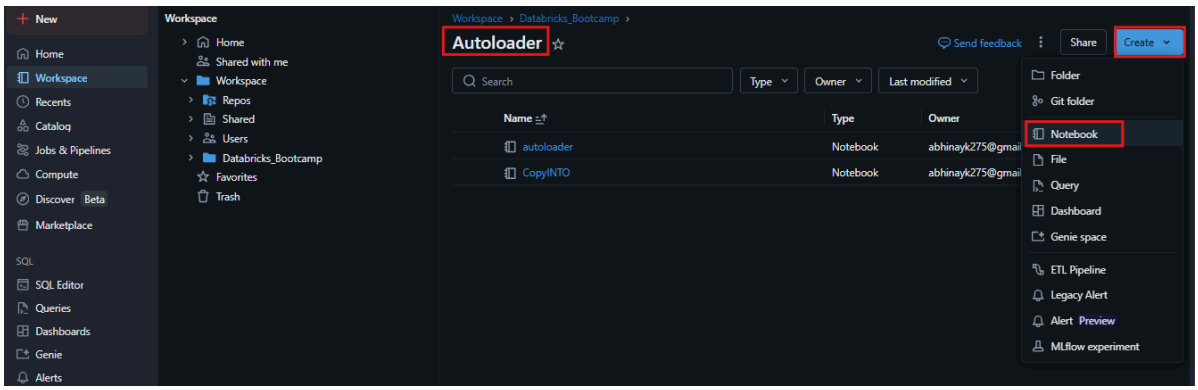
Step 1: Navigate to Workspace

- Go to **Workspace**
- Open **Databricks Bootcamp**
- Navigate to the **autoloader** folder

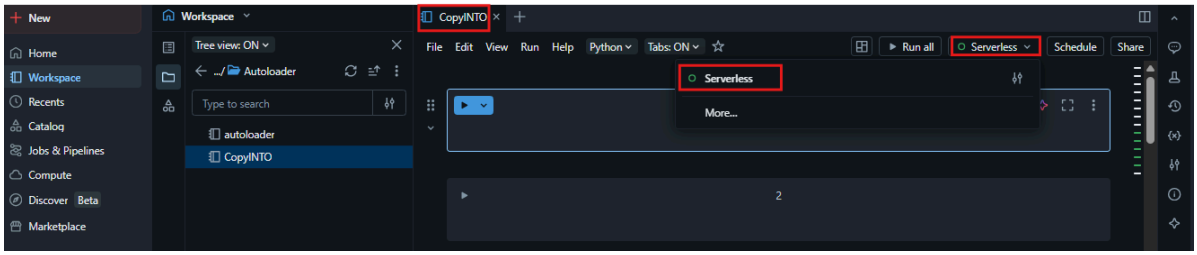


Step 2: Create a Notebook

- Click **Create** → **Notebook**



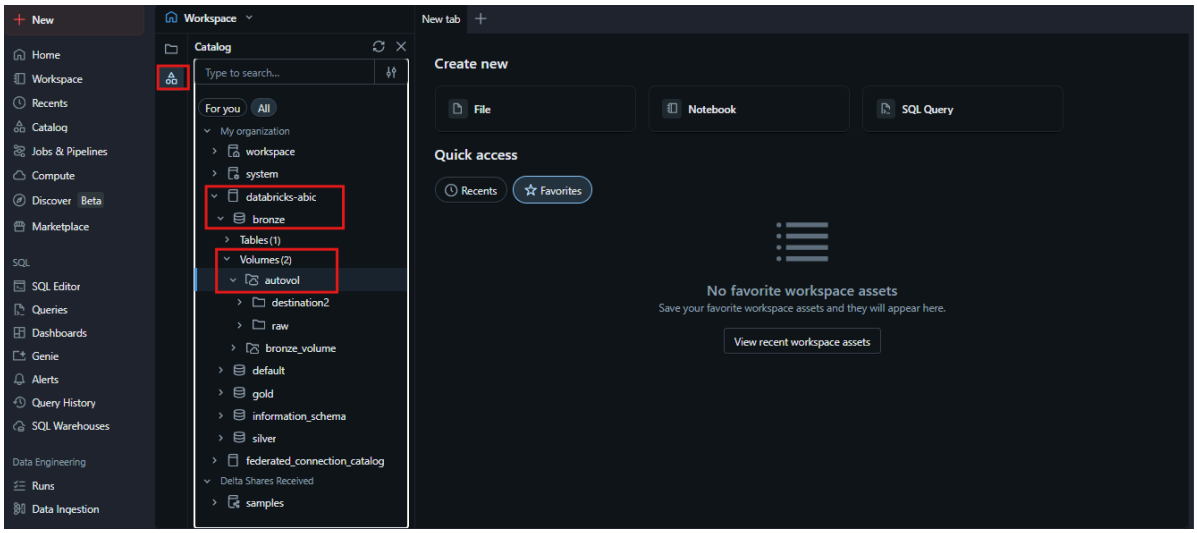
- Provide:
  - Notebook Name: `copy_INTO`
  - Attach **Serverless Compute**



Task 2: Create Directory in Volume

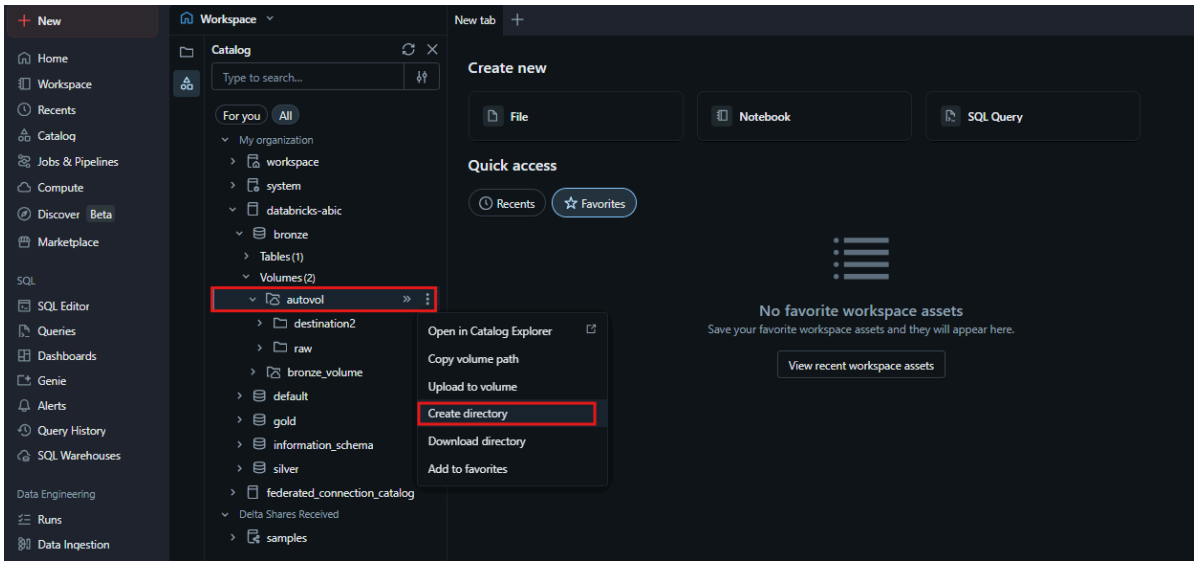
Step 1: Navigate to Volumes

- Go to **Catalog**
- Select **bronze**
- Open **Volumes**
- Navigate to **autovol**



Step 2: Create Folder

- Click **Create Directory**



- Name the folder:

```
copyinto
```

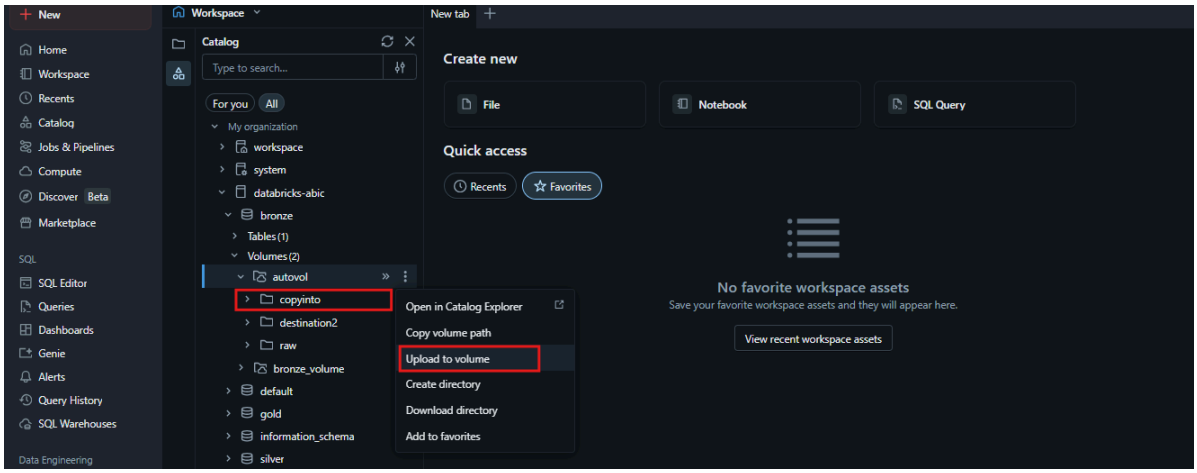
The folder path becomes:

```
/Volumes/databricks-abic/bronze/autovol/copyinto/
```

Task 3: Upload Parquet Files

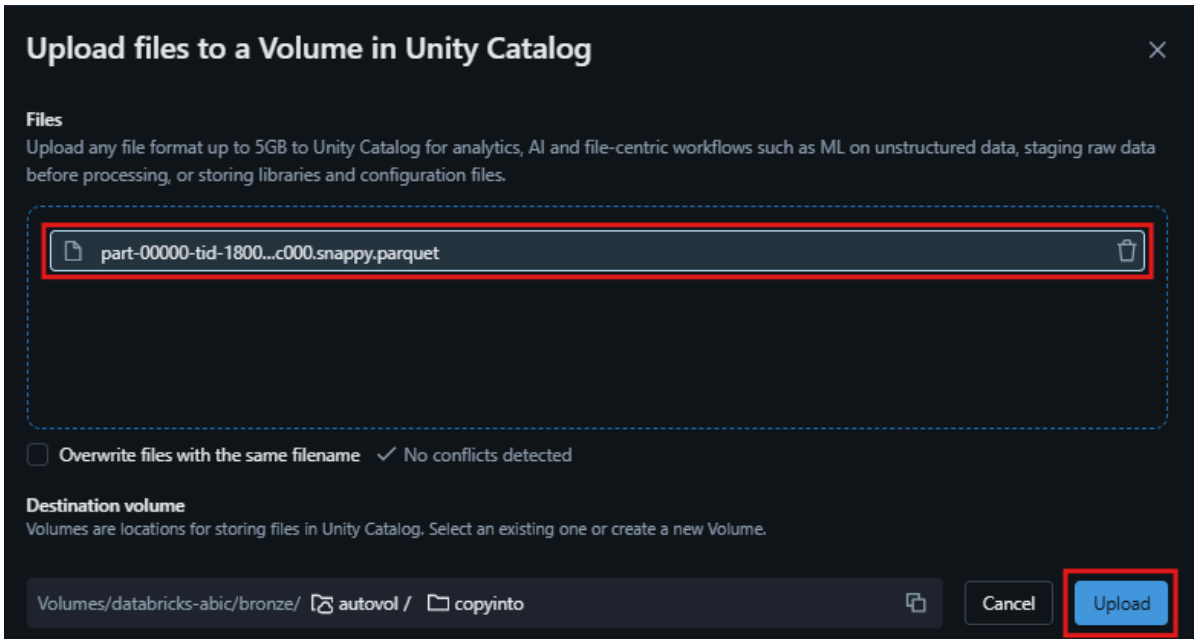
Step 1: Upload First File

- Click **Upload to Volume**



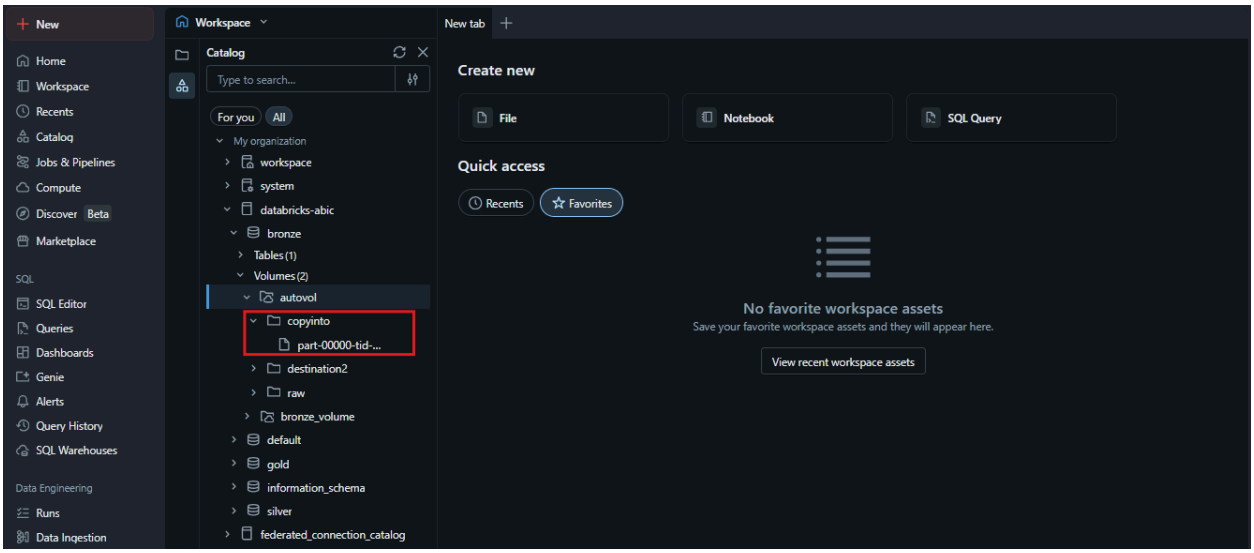
- Select the first Parquet file:

```
orders_1800.......parquet
```



The file will appear in:

```
/Volumes/databricks-abic/bronze/autovol/copy_into/
```

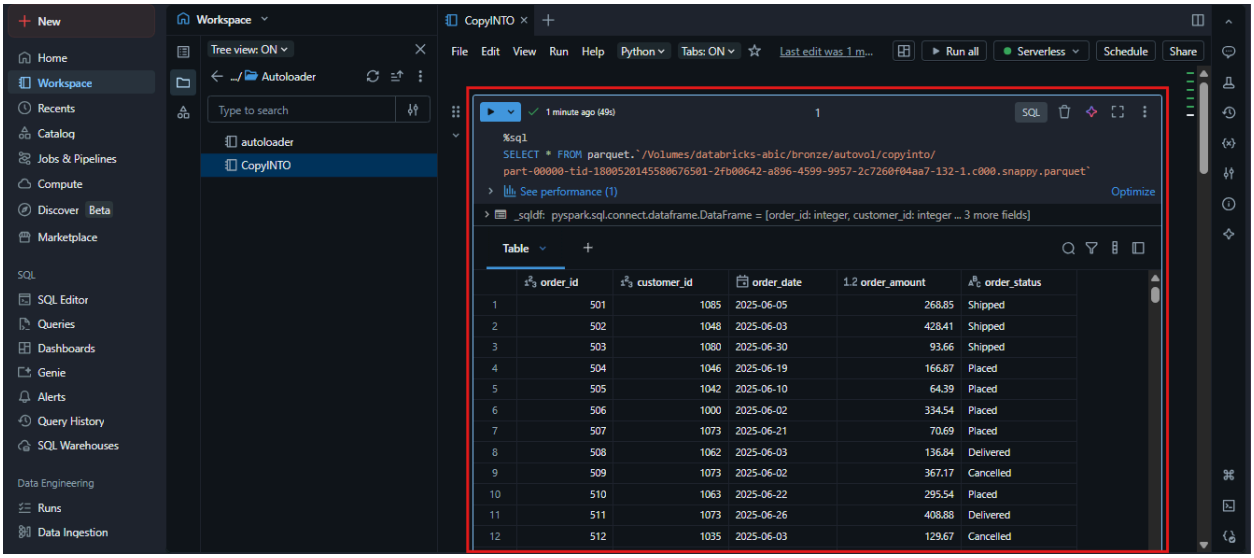


Task 4: View Data from Parquet File

Step 1: Query Parquet File

Run the following SQL command:

```
SELECT *
FROM parquet.`/Volumes/databricks-abic/bronze/autovol/copy_into/orders_1800.parquet`
```



Step 2: Verify Schema

The dataset should contain columns such as:

```
order_id
customer_id
order_date
order_amount
order_status
```

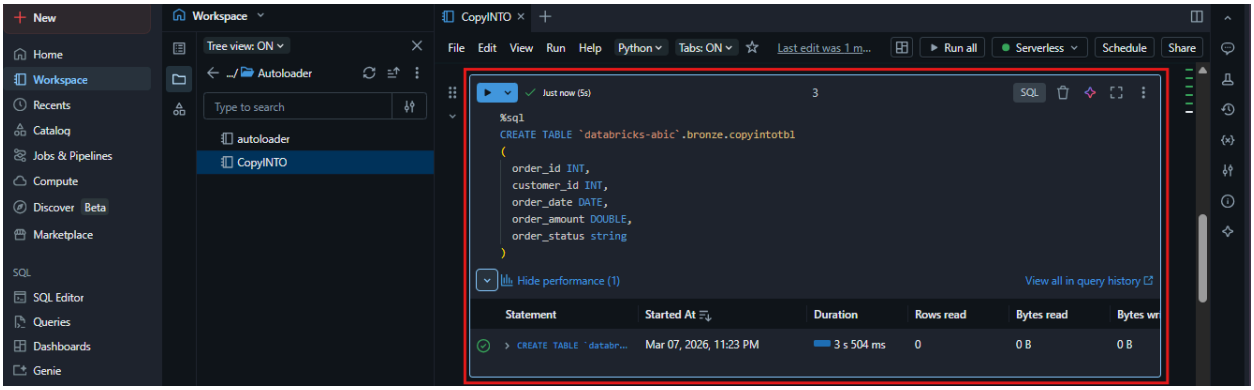
Since Parquet files store schema, the column names will appear correctly.

Task 5: Create Delta Table

Step 1: Create Table

Run the following SQL command:

```
CREATE TABLE `databricks-abic`.bronze.copyintotbl
(
order_id INT,
customer_id INT,
order_date DATE,
order_amount DOUBLE,
order_status STRING
)
```



This creates a **Delta table** that will store the ingested data.

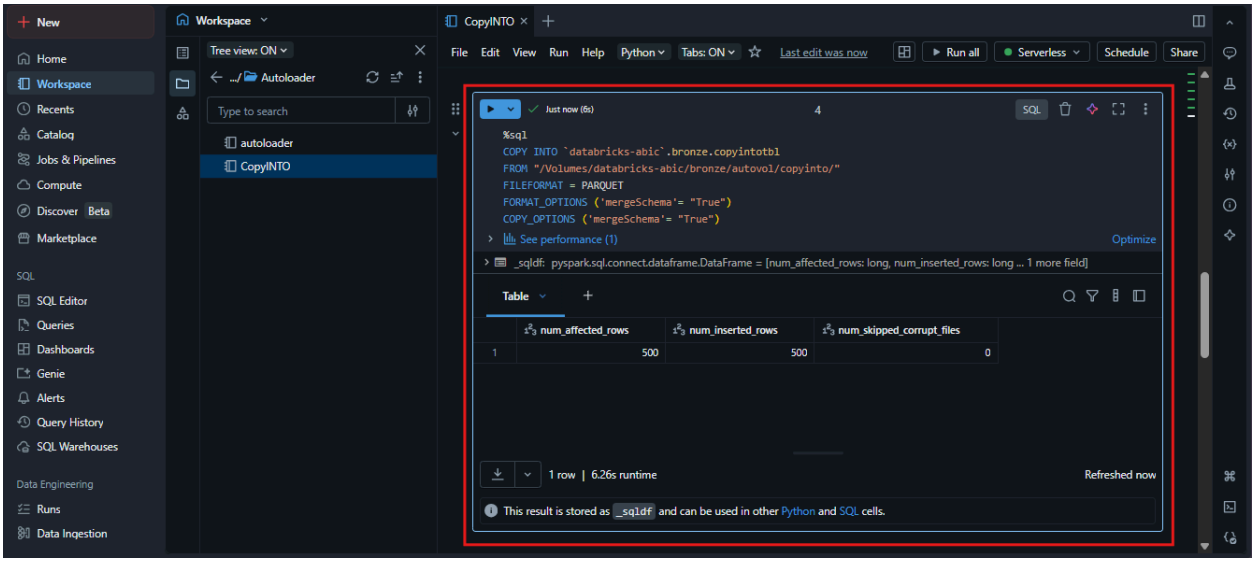
Task 6: Load Data using COPY INTO

Step 1: Execute COPY INTO

Run the following SQL command:

```
COPY INTO `databricks-abic`.bronze.copyintotbl
FROM '/Volumes/databricks-abic/bronze/autovol/copy_into/'
FILEFORMAT = PARQUET
FORMAT_OPTIONS ('mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true')
```

This loads the first Parquet file into the Delta table.

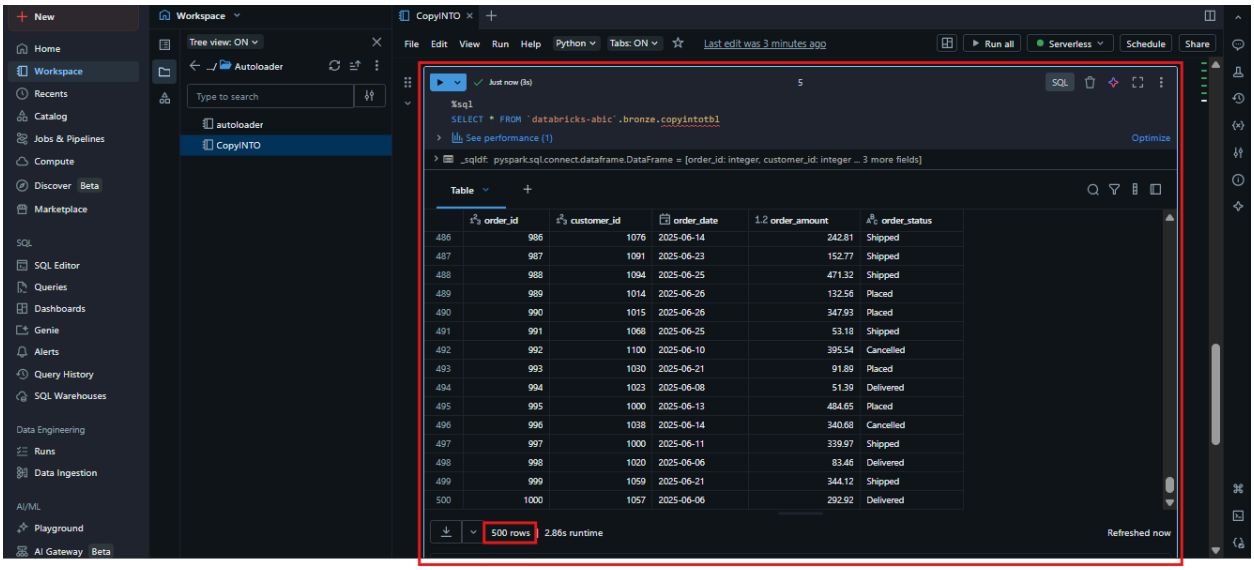


Task 7: Verify Loaded Data

Run the query:

```
SELECT * FROM databricks-abic.bronze.copyintotbl;
```

You should see approximately **500 records** loaded from the first file.



Task 8: Upload Second File

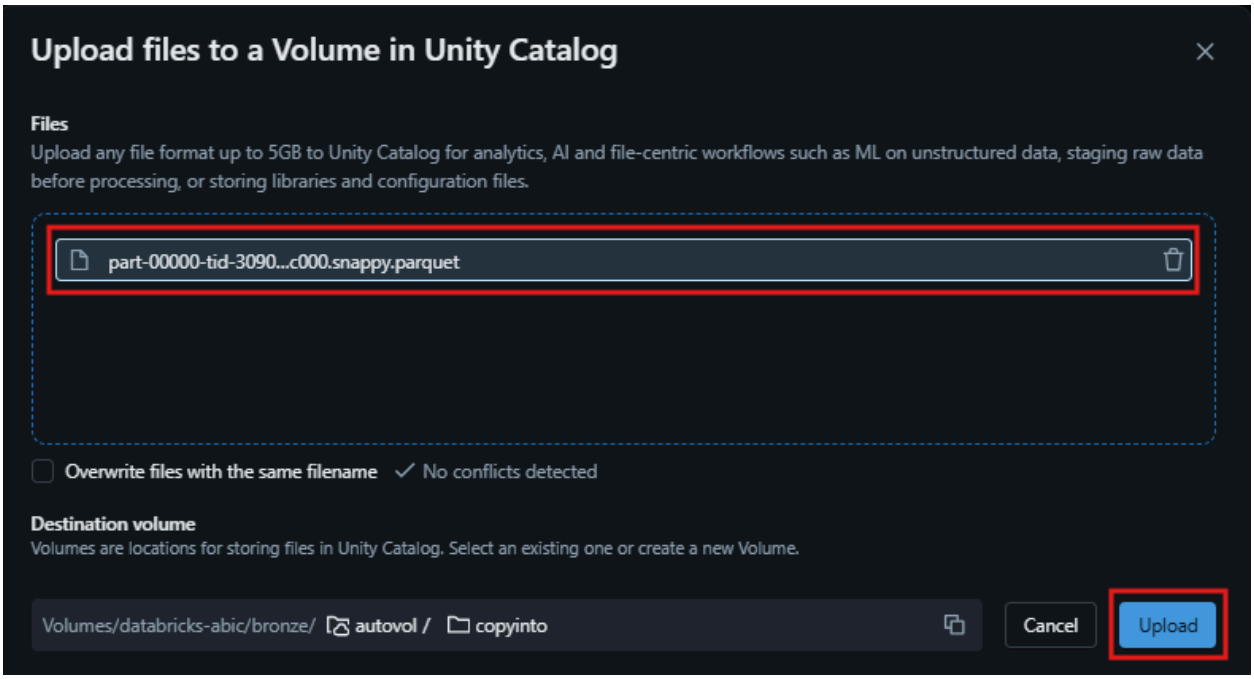
Step 1: Upload Next Dataset

Upload the second Parquet file:

```
orders_3090.parquet
```

to the same folder:

```
/Volumes/databricks-abic/bronze/autovol/copy_into/
```

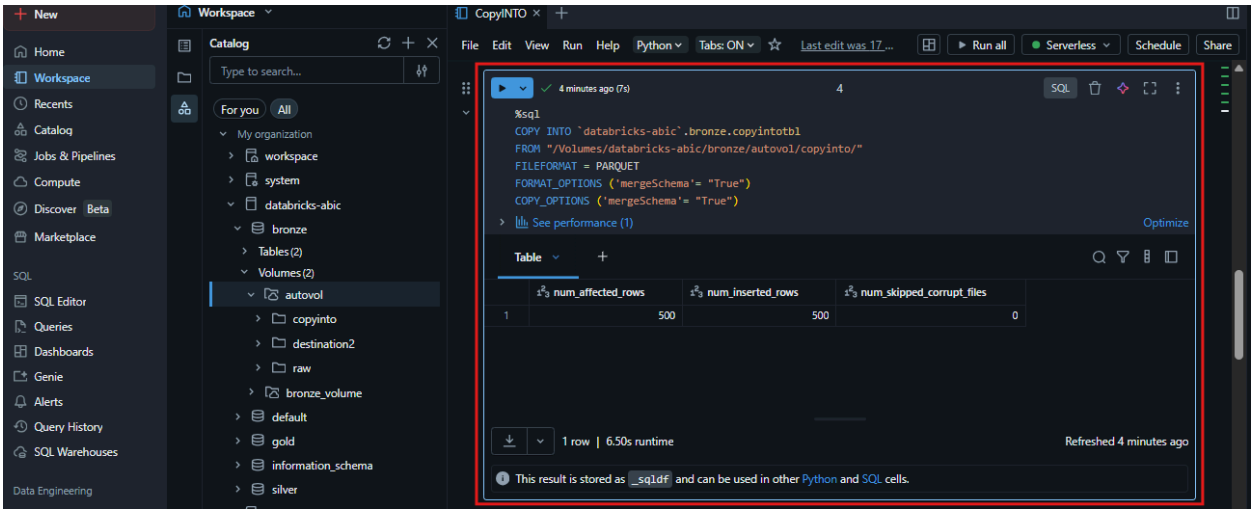


Task 9: Run COPY INTO Again

Execute the same COPY command again:

```
COPY INTO `databricks-abic`.bronze.copyintotbl
FROM '/Volumes/databricks-abic/bronze/autovol/copy_into/'
FILEFORMAT = PARQUET
FORMAT_OPTIONS ('mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true')
```

This will load the second file.



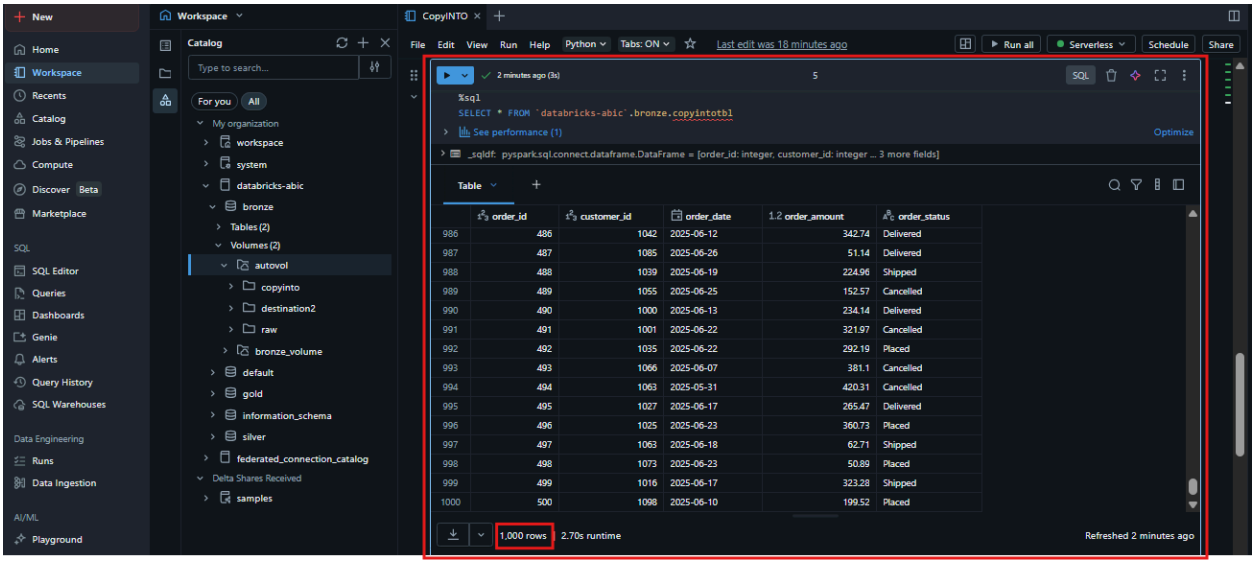
Task 10: Verify Record Count

Run:

```
SELECT COUNT(*) FROM databricks-abic.bronze.copyintotbl;
```

Expected result:

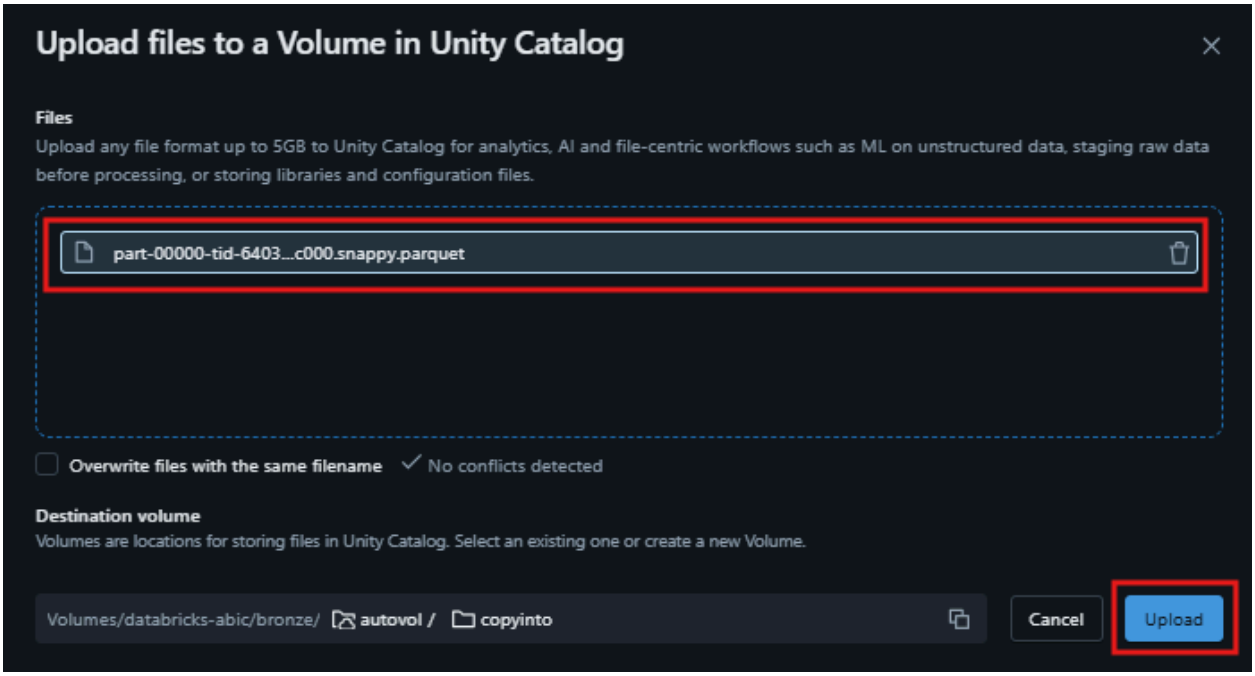
```
1000 records
```



Task 11: Upload Third File (Schema Change)

Upload the third Parquet file:

```
orders_6403.parquet
```



This dataset contains additional columns:

```
discount
payment_method
```

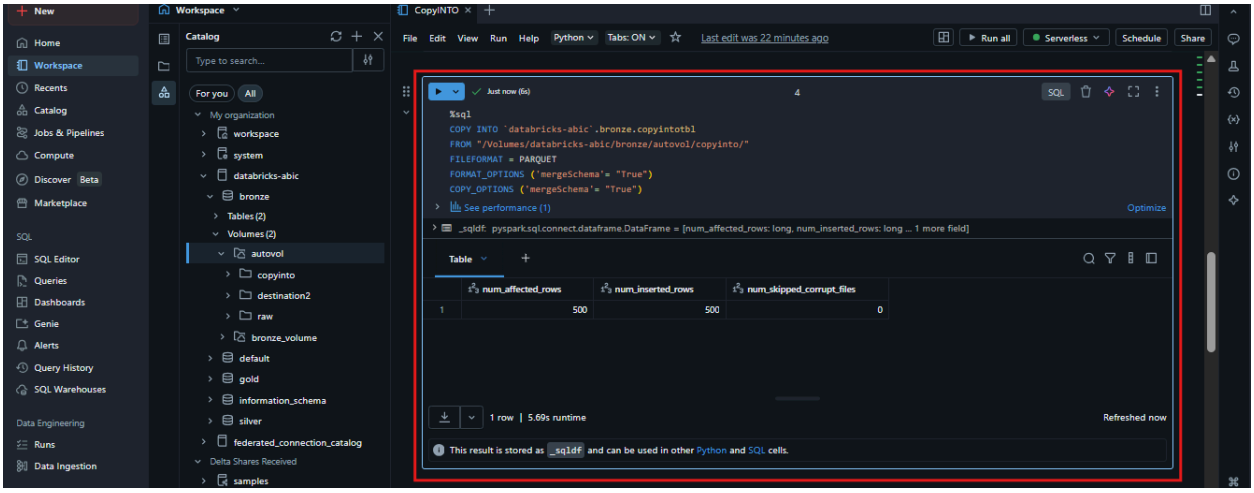
Task 12: Run COPY INTO Again

Execute:

```
COPY INTO databricks.bronze.copyintotbl
FROM '/Volumes/databricks-abic/bronze/autovol/copy_into/'
FILEFORMAT = PARQUET
```

```
FORMAT_OPTIONS ('mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true')
```

The new columns will automatically be added to the table schema.



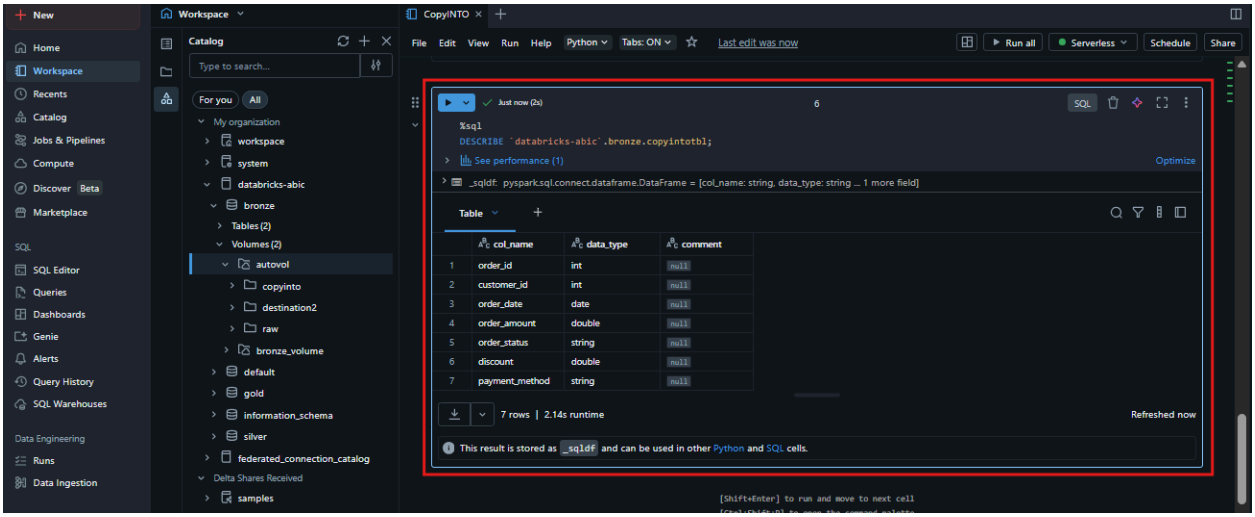
Task 13: Verify Schema Evolution

Run:

```
DESCRIBE `databricks-abic`.bronze.copyintotbl;
```

New columns should appear:

```
discount
payment_method
```



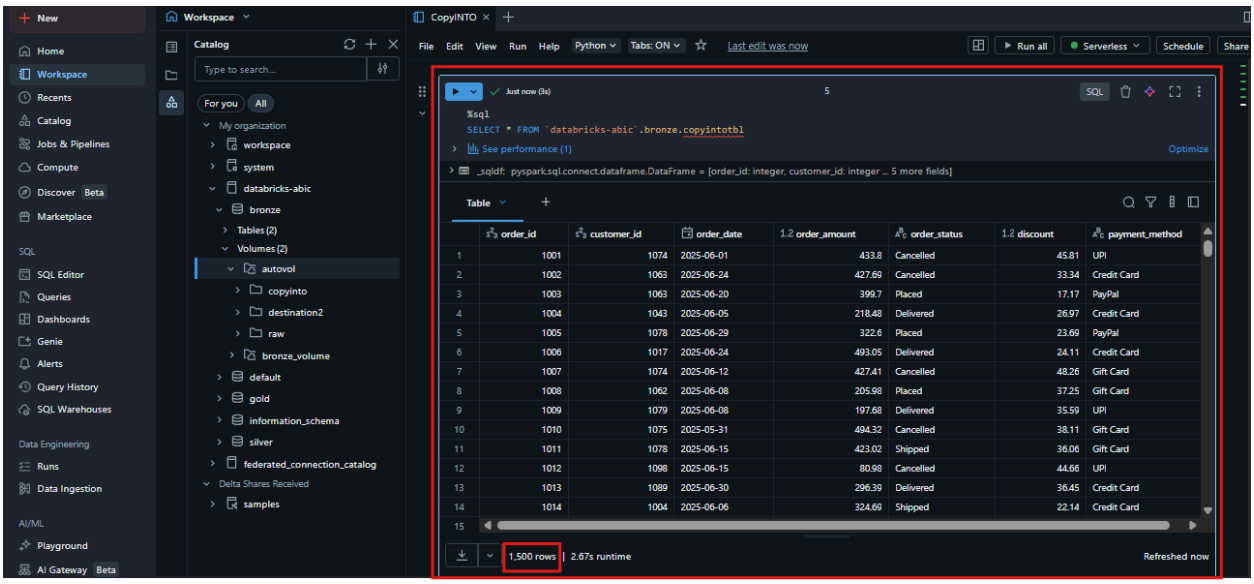
Task 14: Verify Final Data

Run:

```
SELECT * FROM databricks-abic.bronze.copyintotbl;
```

Expected total rows:

```
1500
```



Task 15: Verify Idempotent Behavior

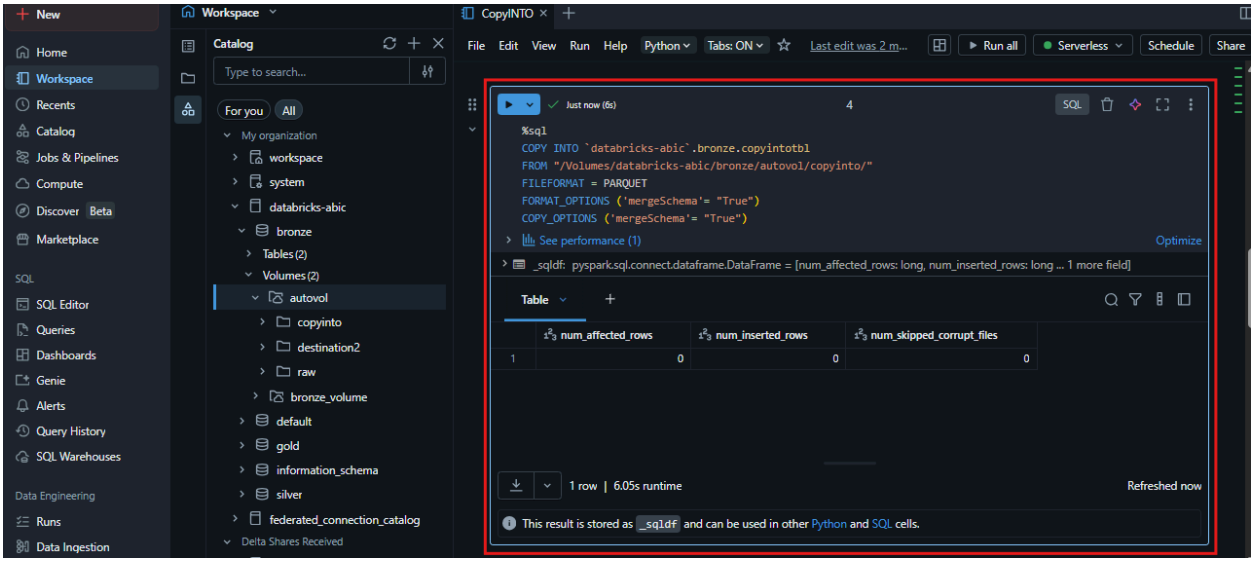
Run the COPY INTO command again:

```
COPY INTO databricks.bronze.copyintotbl
FROM '/Volumes/databricks-abic/bronze/autovol/copy_into/'
FILEFORMAT = PARQUET
FORMAT_OPTIONS ('mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true')
```

Databricks will skip already processed files.

Result:

```
0 files processed
```



Outcome

After completing this lab:

- Created a notebook in Databricks
- Uploaded Parquet files into a volume
- Created a Delta table

- Loaded data using the COPY INTO command
- Implemented schema evolution
- Verified idempotent file loading behavior
- Successfully ingested batch data into Delta Lake